सहायक प्राध्यापक (रसायनशास्त्र) महाराष्ट्र शिक्षण सेवा, गट-अ व अधिव्याख्याता (रसायनशास्त्र) महाराष्ट्र शिक्षण सेवा, गट-ब

(महाविद्यालयीन शाखा)

उच्च व तंत्र शिक्षण विभाग

Assistant Professor (Chemistry) Maharashtra Education Service, Group-A and Lecturer (Chemistry) Maharashtra Education Service, Group-B

(Collegiate Branch)

Higher and Technical Education Department

परीक्षेचे टप्पे: **१)** लेखी परीक्षा - २०० गुण **२)** मुलाखत- ५० गुण

-: परीक्षा योजना :-

विषय व सांकेतांक	माध्यम	प्रश्नसंख्या	गुण	कालावधी	दर्जा	प्रश्नपत्रिकेचे स्वरुप
विषयांशी संबंधीत घटक (११६३)	इंग्रजी	१००	२००	एक तास	पदव्युत्तर पदवी	वस्तुनिष्ठ बहुपर्यायी

अ) नकारात्मक गुणदान -

१) प्रत्येक चुकीच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/ कमी करण्यात येतील.
२) एखाद्या प्रश्नाची एकापेक्षा अधिक उत्तरे दिली असल्यास अथवा ज्या उमेदवाराने उत्तरपत्रिकेत पूर्ण वर्तुळ चिन्हांकित केले नसेल अशा प्रश्नाचे उत्तर चुकीचे समजण्यात येऊन त्या प्रश्नाच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/कमी करण्यात येतील.
३) वरीलप्रमाणे कार्यपध्दतीचा अवलंब करताना एकूण अंतिम गुणांची बेरीज अपूर्णांकात आली तरीही ती अपूर्णांकातच राहील व पुढील कार्यवाही त्याच्या आधारे करण्यात येईल.
४) एखाद्या प्रश्नाचे उत्तर अनुत्तरित असेल तर, अशा प्रकरणी नकारात्मक गुणांची पध्दत लागू असणार नाही.

ब) अंतिम गुणवत्ता यादी ही वस्तुनिष्ठ परीक्षेतील व मुलाखतीतील एकत्रित गुणांवर आधारीत राहील.

-: अभ्यासक्रम :-

विषयांशी संबंधित घटक या विषयांमध्ये खालील घटक व उपघटकांचा समावेश असेल.

अ.क्र.	घटक व उपघटक
	Inorganic Chemistry
1	Chemical periodicity
2	Structure and bonding in homo- and heteronuclear molecules, including shapes of molecules (VSEPR Theory).
3	Concepts of acids and bases, Hard-Soft acid base concept, Non-aqueous solvents.
4	Main group elements and their compounds: Allotropy, synthesis, structure and bonding,
5	Transition elements and coordination compounds: structure, bonding theories, spectral and magnetic properties, reaction mechanisms. Inner transition elements: spectral and magnetic properties, redox chemistry, analytical applications.
6	Organometallic compounds: synthesis, bonding and structure, and reactivity. Organometallics in homogeneous catalysis. Cages and metal clusters.
7	Bioinorganic chemistry: photosystems, porphyrins, metalloenzymes, oxygen transport, electron- transfer reactions; nitrogen fixation, metal complexes in medicine.
8	Characterisation of inorganic compounds by IR, Raman, NMR, EPR, Mössbauer, UV-vis. NQR, MS, electron spectroscopy and microscopic techniques.
9	Nuclear chemistry: nuclear reactions, fission and fusion, radio-analytical techniques and activation analysis.
10	Analytical chemistry- separation, spectroscopic, electro- and thermoanalytical methods.
	Physical Chemistry
11	Basic principles of quantum mechanics: Postulates; operator algebra; exactly- solvable systems: particle-in-a-box, harmonic oscillator and the hydrogen atom, including shapes of atomic orbitals; orbital and spin angular momenta; tunneling.
12	Approximate methods of quantum mechanics: Variational principle; perturbation theory upto second

	order in energy; applications.
13	Atomic structure and spectroscopy; term symbols; many-electron systems and antisymmetry principle.
14	Chemical bonding in diatomics; elementary concepts of MO and VB theories; Huckel theory for conjugated π-electron systems.
15	Molecular spectroscopy: Rotational and vibrational spectra of diatomic molecules; electronic spectra; IR and Raman activities – selection rules; basic principles of magnetic resonance.
16	Chemical thermodynamics: Laws, state and path functions and their applications; thermodynamic description of various types of processes; Maxwell's relations; spontaneity and equilibria; temperature and pressure dependence of thermodynamic quantities; LeChatelier principle; elementary description of phase transitions; phase equilibria and phase rule; thermodynamics of ideal and non-ideal gases, and solutions
17	Statistical thermodynamics: Boltzmann distribution; kinetic theory of gases; partition functions and their relation to thermodynamic quantities – calculations for model systems.
18	Electrochemistry: Nernst equation, redox systems, thermodynamics of electrochemical cells; Electromotive force (EMF) of electrochemical reaction, Debye Huckel theory; activity and activity coefficient, electrolytic conductance – Kohlrausch's law and its applications; ionic equilibria;conductometric and potentiometric titrations.
19	Chemical kinetics: Empirical rate laws and temperature dependence rate of reaction; complex reaction mechanisms steady state approximation; pre-equilibrium approximation, determination of reaction mechanisms; collision and transition state (activated complex) theories ; unimolecular reactions; enzyme kinetics, competitive inhibition, kinetic-salt effects, Chain reaction; homogeneous catalysis; photochemical reactions.
20	Colloids and surfaces: Stability and properties of colloids; isotherms and surface area;heterogeneous catalysis
21	Solid state Chemistry: Crystal structures unit cell, miller indices , X-ray diffraction, Bragg's law and applications; structure factor, atomic scattering factor, Brillouin zones, reciprocal lattice, band structure of solids, electric and magnetic properties of solids
22	Polymer chemistry: Molar masses; kinetics of polymerization.
23	Data analysis: Mean and standard deviation; absolute and relative errors; linear regression;covariance and correlation coefficient.
	Organic Chemistry
24	IUPAC nomenclature of organic molecules including regio- and stereoisomers.
25	Principles of stereochemistry: Configurational and conformational isomerism in acyclic and cyclic compounds; stereogenicity, stereoselectivity, enantio selectivity, diastereo selectivity and asymmetric induction.
26	Aromaticity: Benzenoid and non-benzenoid compounds – generation and reactions.
27	Organic reactive intermediates: Generation, stability and reactivity of carbocations, carbanions, free radicals, carbenes, benzynes and nitrenes.
28	Organic reaction mechanisms involving addition, elimination and substitution reactions with electrophilic, nucleophilic or radical species. Determination of reaction pathways.
29	Common named reactions and rearrangements – applications in organic synthesis.
30	Organic transformations and reagents: Functional group interconversion including oxidations and reductions; common catalysts and reagents (organic, inorganic, organometallic and enzymatic). Chemo, regio and stereoselective transformations.
31	Concepts in organic synthesis: Retrosynthesis, disconnection, synthons, linear and convergent synthesis, umpolung of reactivity and protecting groups.
32	Asymmetric synthesis: Chiral auxiliaries, methods of asymmetric induction – substrate,reagent and catalyst controlled reactions; determination of enantiomeric and diastereomeric excess; enantio-discrimination. Resolution – optical and kinetic.
33	Pericyclic reactions – electrocyclisation, cycloaddition, sigmatropic rearrangements and other related concerted reactions. Principles and applications of photochemical reactions inorganic chemistry.
34	Synthesis and reactivity of common heterocyclic compounds containing one or two heteroatoms (O, N, S).
35	Chemistry of natural products: Carbohydrates, proteins and peptides, fatty acids, nucleic acids, terpenes, steroids and alkaloids. Biogenesis of terpenoids and alkaloids.
36	Structure determination of organic compounds by IR, UV-Vis, 1H & 13C NMR and Mass spectroscopic techniques.

दिनांक – २१ जुलै, २०२५

अवर सचिव
महाराष्ट्र लोकसेवा आयोग